SAMUEL BONNEY

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Samuel Bonney |
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Education

- **Master of Science, Applied Statistics** *Aug 2025 – 2027*
  Illinois State University Normal, IL
  GPA: 3.87/4.00
- **Bachelor of Arts, Mathematics and Statistics** *Sep 2019 – Sep 2023*
  University of Ghana Accra, Ghana
  GPA: 3.64/4.00

Teaching & Work Experience

- **Graduate Teaching Assistant**
  Department of Mathematics, Illinois State University Aug 2025 – Present
  - Facilitate weekly discussion sessions and computational labs, utilizing active learning strategies to reinforce statistical concepts and software proficiency for undergraduate students.
  - Evaluate assignments and examinations with detailed pedagogical feedback, ensuring consistent grading standards while monitoring student academic progress.
- **Teaching and Research Assistant (Statistical Consulting Unit)**
  Department of Statistics and Actuarial Science, University of Ghana Sep 2023 – Aug 2024
  - Coordinated tutorial sessions and comprehensive reviews for undergraduate and graduate cohorts, clarifying complex theoretical frameworks to enhance student retention.
  - Contributed to faculty research within the consulting unit by conducting rigorous data cleaning, literature reviews, and exploratory data analysis (EDA) to support study design.
- **Enumerator and Data Analyst**
  Ghana Baseline Evaluation Survey Jan 2024 – Feb 2024
  - Executed field data collection protocols across multiple sampling units, implementing strict data quality assurance measures to ensure accuracy and minimize entry errors.
  - Synthesized survey data using statistical software to generate empirical insights, directly supporting the monitoring and evaluation of program effectiveness.
- **Data Analyst Intern**
  Perception Management International Limited Sep 2022 – Dec 2022
  - Built pricing models using SQL and Excel for 50+ due diligence engagements across 9 sectors including finance, energy, and telecommunications.
  - Designed automated Power BI dashboards that reduced manual reporting time by 50% (from 8 to 4 hours weekly).
  - Contributed to 6 successful client proposals through data-driven market research.

RESEARCH EXPERIENCE

- **Graduate Thesis** *Aug 2026 – Present*
  Nonparametric Tilted Density Estimation and Bickel–Rosenblatt Goodness-of-Fit Testing
  - Investigating tilted kernel density estimators with cross-validation-based bandwidth and probability-weight selection, using simulation studies to evaluate asymptotic behavior, empirical test size and power, and performance relative to conventional kernel density estimators and Bickel–Rosenblatt goodness-of-fit procedures.
- **Independent Research Project** *May 2026*
  County-Level Lung Cancer Mortality in the United States
  - Integrated CDC SVI, CDC PLACES, and NCI data for 2,536 U.S. counties and applied robust OLS, quantile regression, Moran's I, LISA, spatial error modeling, Random Forest, XGBoost/SHAP, and mediation analysis, identifying 232 unexplained high-mortality spatial clusters and estimating that COPD mediated 50.8% of the county-level smoking–mortality association.

- **Independent Research Project** *March 2026*

  County-Level Food Insecurity and Chronic Disease Burden in the United States
  - Analyzed 2,299 U.S. counties using food insecurity, social-needs, and chronic-disease data; constructed a standardized disease-burden index and applied multivariable regression, sensitivity analysis, spatial autocorrelation, and geographic mapping to identify 408 counties with jointly high food insecurity and chronic disease burden.

- **Large Language Model Project** *Jan 2026 – Apr 2026*

  BioMed-Scribe: Medical Specialty Classification from Clinical Conversations
  - Fine-tuned GPT-2 Medium on 13,492 clinical conversations using two-block parameter-efficient fine-tuning (25.2M trainable parameters; 7.1% of 354.8M total), achieving 79.1% test accuracy and 0.799 macro-F1 across Cardiology, Neurology, Orthopedics, and General Medicine, with t-SNE embedding analysis and deployment through an interactive Gradio application.

- **Academic Project** *Jan 2026 – May 2026*

  Statistical Analysis of Diabetes Risk Factors Among American Adults
  - Analyzed 253,680 CDC BRFSS observations using SAS and R, applying data transformation, chi-square tests, factorial ANOVA, multiple and logistic regression, model selection, diagnostics, and sex-stratified analyses to identify clinical, behavioral, demographic, and socioeconomic factors associated with diabetes-related outcomes.

- **Academic Project** *Oct 2025*

  Modeling Life Expectancy in Africa Using Socioeconomic and Health Indicators
  - Implemented a multiple linear regression model using socioeconomic data to predict life expectancy; the final model achieved an Adjusted R^2 of 0.98, successfully identifying Adult Mortality and HIV incidence as the most significant drivers.

- **Research Project** *Jan 2024 – Aug 2024*

  Modeling The Prevalence of Malaria Among [pregnant women] in Ghana Using Explainable Machine Learning Techniques
  - Applied machine learning algorithms to Demographic and Health Survey data to predict malaria risk; the K-Nearest Neighbor model yielded the highest performance (88% AUC), successfully ranking factors like age and anaemia status by importance

- **Undergraduate Thesis** *Jan 2023 – Sep 2023*

  Application of Machine Learning Models to Identify the Risk Factors of Incidence of Malaria Among Children Under Five Years of Age in Ghana
  - Evaluated five machine learning algorithms, including Random Forest and Neural Networks, to predict malaria incidence using the 2019 Ghana Malaria Indicator Survey, successfully identifying key risk determinants for pregnant women.

- **Academic Project** *Jan 2022*

  Fractal Pattern Generation via Iterated Function Systems
  - Developed a Python program that applied affine transformations to generate fractal patterns by iteratively mapping geometric shapes using matrix operations and translation vectors, demonstrating practical understanding of linear algebra, NumPy computation, and visualisation of iterated function systems.

Skills

- **Statistical Analysis & Modeling:** R, SPSS, SAS, STATA
- **Programming & Database Management:** Python, SQL
- **Technical Documentation & Version Control:** LaTeX, R Markdown, Microsoft Office Suite, Git

Honors & Awards

- **Outstanding Master's Student,** Illinois State University, 2026
- **Kenneth Berk Scholarship,** Illinois State University, 2026
- **Joint Statistical Meeting Diversity Workshop and Mentoring program Funding,** JSM DWMP, 2026
- **Fostering Diversity in Biostatistics Workshop Funding,** ENAR, 2025
- **Winner, Statistics and Actuarial Science Quiz Competition,** University of Ghana, 2023
- **Winner, Mathematics Society Quiz Competition,** University of Ghana, 2023

Conferences & Workshops

Joint Statistical Meetings (JSM) and Diversity Mentoring Program (DMP), ASA, Boston, MA *Aug 2026*

ENAR Fostering Diversity in Biostatistics Workshop, Indiana, IN *Mar 2026*

Presentations

County-Level Food Insecurity and Chronic Disease Burden in the United States: Identifying Priority Counties for Intervention, Research Presentation, Illinois State University *May 2026*

A Statistical Analysis of Diabetes Risk Factors Among American Adults Using SAS and R: A Sex-Based Comparative Approach, Class Presentation, Illinois State University *May 2026*

County-Level Lung Cancer Mortality in the United States: Social Vulnerability, Behavioral Risk Factors, Spatial Clustering, and Unexplained Mortality Hotspots, Research Presentation, Illinois State University *March 2026*

A Statistical Analysis of Cardiovascular Disease Risk Factors Using Minitab: A Gender-Based Comparative Approach, Class Presentation, Illinois State University *Dec 2025*

Application of Machine Learning Models to Identify the Risk Factors of Incidence of Malaria Among Children Under five years in Ghana, Departmental Seminar, University of Ghana *Nov 2023*

Professional Memberships

American Mathematical Society *Oct 2025 – Present*

Institute of Mathematical Statistics *Sept 2025 – Present*

Certifications & Courses

SQL Fundamentals, DataCamp *Feb 2026*

Functions for Manipulating Data in PostgreSQL, DataCamp *Feb 2026*

Working with Dates and Times in R, DataCamp *Jan 2026*

Intermediate SQL, DataCamp *Oct 2025*

Leadership & Professional Service

Treasurer, Ghanaian Students Association *2026 – Present*
Illinois State University, Normal, IL

- Oversee budgeting, financial recordkeeping, reimbursements, and expenditure tracking while collaborating with executive members on event planning, resource allocation, and responsible financial decision-making.

MathEye Impact *2021 – Present*
Co-Founder / Vice President

- Co-founded and directed a non-profit school for high-achieving, low-income students from across Ghana, mainly the Central Region.
- Guided inaugural class to best-in-country academic results on Cambridge International Exams, with students earning A & A* grades on over 70% of exams taken, and 5 students finishing #1 in Ghana in their subject.
- Led all aspects of a 275-student boarding school for high-achieving, low-income students.

Solution Space Member *2020 – 2023*
The Mathematics Society, University of Ghana

- Guided students in understanding and approaching complex mathematical problems by providing clear explanations and strategies, fostering independent problem-solving skills.